

MINERVA CryoCell P&ID v3 — A3 Printing Guide

All v3 sheets are authored at true **A3 landscape — 420 × 297 mm** with a `viewBox` of `0 0 1587.273 1122.430` user units (≈ 3.779 units/mm). The PDFs are already A3; the SVGs declare `width="420mm" height="297mm"`.

Recommended output

Use	File	Notes
Colour review / plotting	*_STANDARD.pdf	Full colour, process emphasised
Control-room / signal review	*_CONTROL-CENTRIC.pdf	Process greyed back, signals emphasised
Black-and-white plotting	*_STANDARD_MONO.pdf / *_CONTROL-CENTRIC_MONO.pdf	Line-weight + dash differentiation only

Printer settings

- **Paper:** A3 (420 × 297 mm), **landscape**.
- **Scaling:** 100 % / Actual size — do **not** “fit to page” (it rescales line weights and defeats the calibrated 0.25–1.0 mm hierarchy).
- **Margins:** borderless or ≤ 6 mm; the sheet already carries a 6 mm frame margin.
- **Colour:** use the `_MONO` files for B/W printers rather than letting the driver desaturate — the mono files are designed to read without colour.

Text legibility (verified minimums @ A3)

Element	Target	v3 setting
Main equipment / line tags	≥ 2.5 mm	2.6 mm
Instrument bubble text	2.0 mm	~ 1.85 –2.0 mm
Line / note callouts	2.2 mm	2.2 mm
Legend text	1.8 mm	1.8 mm

Line-weight hierarchy (at 100 %)

- Primary cryo trunk **1.0 mm** → branch **0.7 mm**.
- Secondary (DI water) **0.5 mm**.
- Out-of-scope services **0.35 mm** (dashed).
- All instrument signals **0.25 mm** (pneumatic dash+//, electric dotted, hydraulic dash-dot).

Regenerating PDFs from SVG

```
cd pid_project
python3 -c
"import glob,cairosvg; [cairosvg.svg2pdf(url=s, write_to=s[:-4]+'.pdf') for s in
glob.glob('output_v3/*//*.svg')]"
```

`cairosvg` preserves the 420 × 297 mm page size ($\approx 1190.55 \times 841.89$ pt).

Tip — declutter before printing

Use a **default view** (see `DEFAULT_VIEWS_GUIDE.md`) to hide layers you do not need, and turn the `16_Legend_TOGGLEABLE` layer **on** if the print is going to someone unfamiliar with the symbol set.